MILITARY SPECIFICATION

REPAIR KIT, SEALS, EXPANDABLE AND

MULTI-PURPOSE SHELTER (MUST)

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of seal repair kit for the MUST expandable and multi-purpose shelters.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

5-4-3598 - Repair Kit, Seal, Expandable and Multi-Purpose Shelter (MUST)
5-4-3599 - Storage Box, Repair Kit (MUST)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a complete sample of the seal repair kit for inspection and approval (see 4.3 and 6.3).

3.2 Component parts. The seal repair kit shall consist of the items listed below, in the stated quantities and conforming to the applicable referenced material or source.

Item No.	Description	Quantity
1	Seal, bulb and tail, rubber	300 feet
2	Seal, extruded, olive drab	36 feet
3	Seal, extruded, olive drab	100 feet
4	Seal, extruded, olive drab	100 feet
5	Seal, extruded, olive drab	25 feet
6	Seal, extruded, black	3 feet
7	Seal, 1 inch wide by 1/16 inch thick	12 feet
8	Seal, 1/2 inch wide by 1/16 inch thick	60 feet
9	Seal, 5/8 inch wide by 1/4 inch thick	20 feet
10	Seal, 1-1/8 inches wide by 1/16 inch thick	8 feet
11	Seal, 1/2 inch wide by 1/4 inch thick, black	10 feet
12	Seal, 4 inches wide by 1/16 inch thick	40 feet
13	Seal, 5/8 inch wide by 3/16 inch thick	7 feet
14	Seal, 2 inches wide by 3/16 inch thick	12 feet
15	Seal, 1/4 inch wide by 1/4 inch thick	6 feet
16	Seal, 3/4 inch by 3/4 inch	6 feet

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

Item No.	Description	Quantity
17	Rivet, domed, CLE, 1/8 inch dia. by 3/16 inch long	500
18	Rivet, domed, CLE, 1/8 inch dia. by 1/4 inch long	500
19	Rivet, 120° CSKH, CLE, 1/8 inch dia. by 3/16 inch long	100
20	Rivet, 120° CSKH, CLE, 1/8 inch dia. by 5/16 inch long	100
21	Rivet, 120° CSKH, CLE, 3/16 inch dia. by 3/8 inch long	100
22	Rivet, 120° CSKH, CLE, 3/16 inch dia. by 1/4 inch long	100
23	Rivet, domed, CLE, 3/16 inch dia. by 1/4 inch long	500
24	Sealant, rivet	1 gal.
25	Adhesive, seal, 2 parts: 2 qts. part A, 2 qts. part B	1 gal.
26	Solvent, MEK, Technical (requisitioned on as needed basis)	1 gal.
27	Cloth, abrasive, coated, alum-oxide or sil-carbide 9 inch by 11 inch sheets, type I, class 1	12 sheets
28	Brush, varnish, 1 inch width	12
29	Tool, installing, rivet	1
30	Towel, wiper, paper, industrial and institutional, type I	4 boxes
31	Box, storage, repair kit (MUST)	1

3.3 Design and construction. The design and construction of the repair kit shall conform to the requirements specified herein, on the applicable drawings and referenced documents.

3.4 Markings. Markings for the box shall conform to the requirements specified on Drawing 5-4-3599.

3.5 Finish. All components shall be finished in accordance with the applicable requirements of referenced specifications and drawings. Commercial items shall be finished in accordance with the manufacturer's standard practice.

3.6 Workmanship. Metal components shall be free from sharp edges, slivers or burrs. Wood shall not be split nor shall nail points protrude through wood. Enameling shall be dry, free from runs, wrinkles, streaks, area of no film, and grit or coarse particles imbedded under finish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects listed in table I and 4.4.2.2. The presence of any defects shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The lot shall consist of all completely fabricated seal repair kits offered for inspection at one time. The sample unit shall be one completely fabricated seal repair kit.

4.4.2.1 Visual inspection. The seal repair kits shall be examined for defects in table I. The inspection level shall be level II and the acceptable quality level (AQL) expressed in terms of defects per hundred units shall be 2.5 for major defects, and 6.5 for total defects.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:28Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects for visual characteristics

Examine	Defect	Classification Major	Minor
Finish	Primer and enameling not as specified.	X	
	Not color specified.		X
	Area of no film.		X
	Not continuous or is not smooth.		X
	Contains runs, drops, or streaks.		X
	Foreign matter imbedded		X
	Evidence of peeling, flaking, or blistering.		X
Construction and workmanship	Component missing.	X	
	Any burrs, slivers, splinters, or rough or sharp edges.	X	
	Component broken, cracked or malformed.	X	
Marking	Omitted, incomplete, illegible, incorrect, or wrong size or location.		X

4.4.2.2 Dimensional examination. Examination shall be made of boxes for dimensions specified. Any nonconformance shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking requirements are in compliance with section 5. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully packaged except that it need not be closed. Examination for the closure defects listed below shall be made on a shipping container fully packaged. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

TABLE II. Packaging inspection

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping or tape banding, and inadequate stapling. Bulged or distorted container.
Contents	Not as specified.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. All components of the repair kit shall be placed inside the storage box. The storage box lid shall be closed and the hasp secured.

5.2 Packing. Packing shall be level A.

5.2.1 Level A packing. Each repair kit, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container constructed in accordance with style FOL, grade V3c, V3s or V4s of PPP-B-636. Each shipping container shall be closed in accordance with method III and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be as specified in 4.4.3.

5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This seals repair kit is provided to make repairs to the expandable and multi-purpose shelters (MUST) in the field.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required (see 3.1, 4.3 and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under appropriate provisions of Armed Services Procurement Regulations ASPR 7-104.55. The first article should be a preproduction sample consisting of one completed seal repair kit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 5410-A246

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-44032(GL) Repair Kit, Seals, Expandable and Multi-Purpose Shelter (MUST)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760
OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-ES

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314



Commander
US Army Natick Research and Development Command
ATTN: DRDNA-ES
Natick, MA 01760

NATICK OP 1 AUG 78
FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:28Z
Check the source to verify that this is the current version before use.